Telco Customer Churn Prediction

Case Study | Anthonia Ozobialu | Data Analyst

Business Problem

Customer churn directly erodes recurring revenue for telecom providers, and acquiring a new customer costs significantly more than retaining an existing one. This project analyzes a telecom customer base to identify which customers are most likely to churn and why, giving the business a way to act on retention before customers leave rather than after.

Data & Methodology

The dataset covers 7,043 customers (7,032 after cleaning) with demographic, account, service, and billing features, sourced from the widely-used Kaggle Telco Customer Churn dataset. The workflow included:

- Exploratory data analysis of churn patterns across contract type, tenure, payment method, and internet service
- Feature engineering and preprocessing (encoding, scaling, handling missing values)
- Model comparison: Logistic Regression, Random Forest, and XGBoost
- Evaluation on Recall, F1, and AUC-ROC — best result: Logistic Regression at 0.845 AUC-ROC
- SHAP analysis for model explainability and feature importance

Key Findings

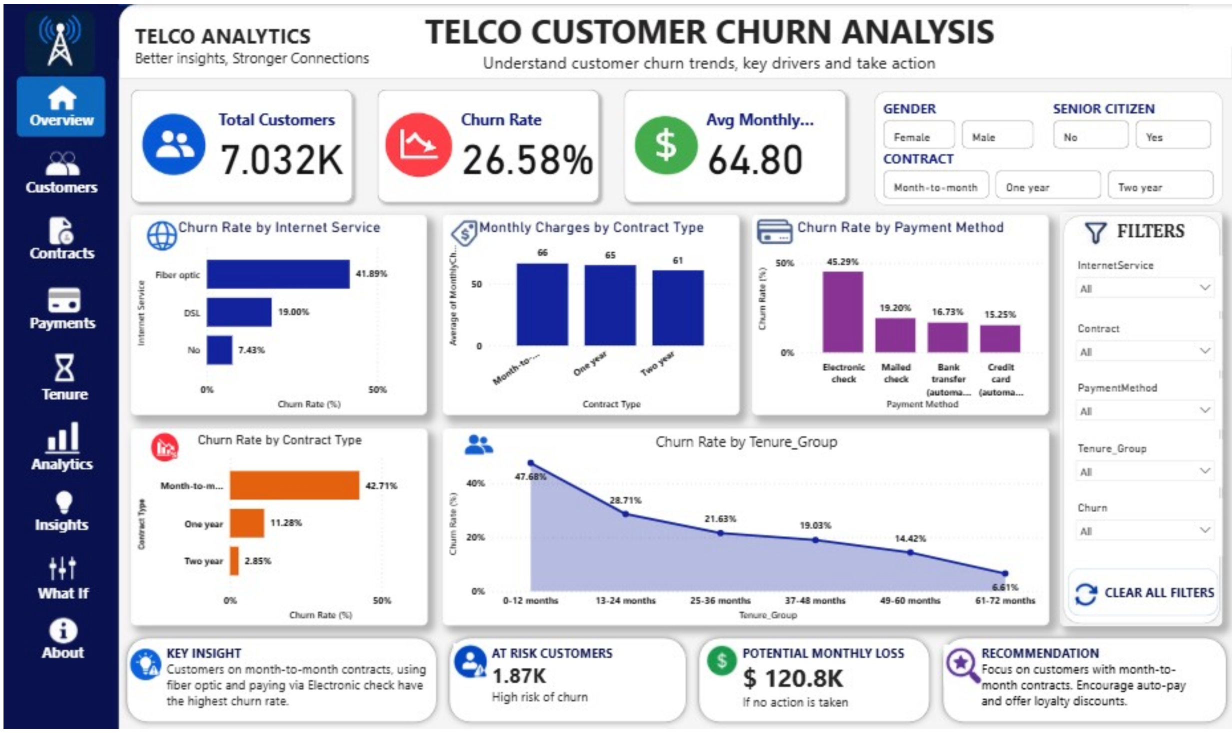
- Customers on month-to-month contracts, fiber optic internet, and electronic check payment form the highest-risk churn cluster
- Month-to-month contracts see a 42.71% churn rate, versus just 2.85% for two-year contracts
- Churn risk drops sharply once a customer passes the 12-month tenure mark

Business Impact

Metric	Value
Customers flagged high-risk	1,870
Estimated revenue at risk / month	$120,800
Recommended action	Auto-pay incentives + loyalty discounts for month-to-month segment

Interactive Dashboard

A Power BI dashboard was built to make these insights explorable by segment, with filters for contract type, internet service, payment method, and tenure group.



Telco Customer Churn Analysis dashboard — overview page

GitHub Repository: github.com/AnthoniaOzobialu/AnthoniaOzobialu-data-portfolio/tree/main/churn-prediction

Dataset Source: kaggle.com/datasets/blastchar/telco-customer-churn